



MMLoRA: Multitask Memory Parameter-Efficient Fine-Tuning for Multimodal SER

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Abstract

The differences in emotional expression present significant challenges for the development of effective emotion recognition systems. Although large language models (LLMs) have demonstrated strong performance, their generalization in emotion recognition tasks is often limited by variations in individual emotional expression, which remain inadequately addressed even by parameter-efficient fine-tuning techniques such as LoRA. To address these challenges, we propose a multitask memory parameter-efficient fine-tuning method (MMLoRA) that enhances multimodal SER by incorporating gender as an auxiliary task. The method leverages shared LoRA experts to facilitate information exchange between tasks and employs a mixture of LoRA experts to process task-specific information. Additionally, a memory mechanism propagates task-specific information across layers. Experimental results demonstrate that the proposed MMLoRA significantly improves emotion recognition performance compared to vanilla LoRA.

Index Terms: large language model, LoRA, multimodal, multitask, speech emotion recognition.

1. Introduction

Speech emotion recognition (SER) is essential for improving the effectiveness of human-computer interaction and personalized services, and it plays a pivotal role in advancing emotional intelligence systems [1]. A key challenge in SER lies in the variability of emotional expression across individuals, as the same emotion can be conveyed differently by different people [2, 3]. To address this challenge and enhance cross-individual generalization, researchers have employed multitask learning strategies that account for individual differences, including gender [4, 5], identity [6], and so on, thereby improving the accuracy of emotion recognition.

In recent years, LLMs have achieved significant breakthroughs across various domains [7]. Notable examples include models like LLaMA2 [8] and Qwen [9] in natural language processing (NLP), as well as Multimodal Large Language Models (MLLMs) like Video-LLaMA[10], LLaVA [11] and GPT-4 [12], which extend these advancements to tasks involving multimodal information. These large-scale pre-trained models leverage extensive pre-training to capture intricate patterns in data, demonstrating outstanding performance when fine-tuned on a wide range of downstream tasks. However, as the scale of these models continues to grow, the computational cost associated with comprehensive fine-tuning has become a significant challenge. Low-Rank Adaptation (LoRA) [13] has emerged as an effective approach, freezing pre-trained model weights and in-

roducing trainable low-rank matrices, making it a valuable fine-tuning strategy in resource-constrained settings. This approach has shown promise in enhancing the adaptability of LLMs to specific tasks, such as SER

Despite the success of LoRA in alleviating GPU memory limitations during fine-tuning, its application to multi-task learning scenarios exposes critical limitations [14]. Fundamental challenges stem from the restricted trainable parameter budget and persistent catastrophic forgetting [15], jointly impairing the model's cross-task generalization capabilities. Current efforts in multi-task fine-tuning predominantly focus on resolving inter-task conflicts through architectural innovations. Chen et al. [16] developed a sparse mixture of LoRA experts to enhance the baseline architecture, effectively alleviating data conflicts during multimodal large language model (MLLM) instruction tuning across heterogeneous domains. Similarly, Song et al. [17] devised language-specific LoRA modules for Whisper models to reduce linguistic interference while preserving existing knowledge during new language integration. While these approaches demonstrate progress in task conflict mitigation, they predominantly adopt a parameter-isolation paradigm that enforces single-task specialization during inference. This architectural constraint not only diminishes the synergistic benefits of multi-task learning but also fundamentally restricts downstream tasks from harnessing cross-task knowledge transfer, which is a critical advantage inherent to multi-task frameworks.

Multitask learning strategies that incorporate personalized knowledge have proven effective in Speech Emotion Recognition (SER) for addressing individual differences, thereby improving emotion recognition accuracy. To extend these benefits while overcoming the task-isolation limitations of existing parameter-efficient fine-tuning methods, this paper proposes a novel multitask parameter-efficient fine-tuning method for multimodal SER. In the proposed MMLoRA, we adopt gender recognition and emotion recognition as the multitask setting. Specifically, a universal LoRA expert is introduced to facilitate cross-task information sharing, while task-specific information is captured through a mixture of task-specific LoRA experts. Additionally, a memory mechanism is incorporated, allowing the model to retain and utilize task-specific information across layers, thus enabling simultaneous multitask inference. In our experiments, VideoLLAMA serves as the base multimodal pre-trained language model, though this approach is not limited to this model and can be applied to other open-source large models as well. The main contributions of this paper are summarized as follows:

- We propose MMLoRA, a parameter-efficient multimodal Speech Emotion Recognition (SER) framework that overcomes conventional LoRA's limitations in multitask inter-

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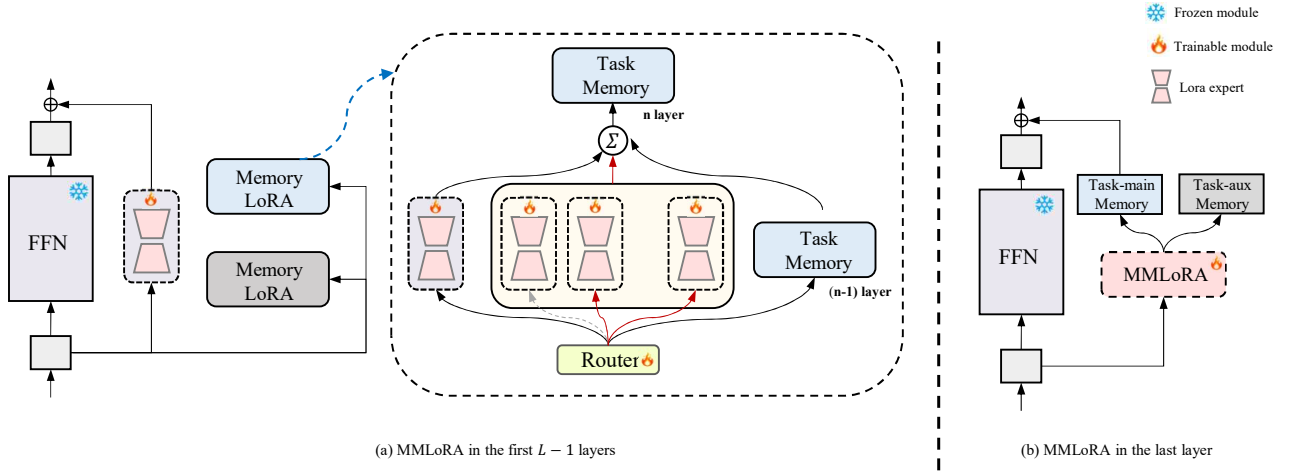


Figure 1: **Overview of the MMLoRA Method.** The MMLoRA module is integrated into the feed-forward network (FFN) of transformer-based blocks. (a) In the first $L - 1$ layers, MMLoRA employs a task-specific memory mechanism to enable the model to retain and propagate the information of each task across layers. (b) In the last layer, the memory from each task (i.e., the main task and the auxiliary task) is added to the output to generate task-specific results.

action. By integrating a universal LoRA expert with task-specific mixtures and a memory mechanism, MMLoRA explicitly leverages auxiliary tasks (e.g., gender recognition) to mitigate gender bias and enhance emotion recognition accuracy.

- Experiments validate that MMLoRA surpasses baseline methods in SER performance through its multi-task learning design, while maintaining parameter efficiency. This framework provides a generalizable paradigm for deploying lightweight multitask systems in resource-constrained scenarios.

2. Method

In this paper, we focus on enhancing the performance of efficient parameter fine-tuning methods in MLLMs for multimodal SER through multitask learning. The multitask setting is considered as a gender-assisted emotion recognition task in this paper. We validate the effectiveness of the proposed MMLoRA method within an open-source multimodal LLM, Video-LLaMA. Section 3.1 provides a detailed explanation of the proposed MMLoRA architecture, while Section 3.2 outlines the fine-tuning process for the gender-assisted multimodal SER task within Video-LLaMA model.

2.1. Preliminaries

LoRA was introduced as an effective technique for fine-tuning pre-trained LLMs with reduced computational cost. It works by freezing the original weights and introducing trainable low-rank decomposition matrices into specific layers. This approach significantly reduces the number of trainable parameters, making it more computationally efficient without sacrificing performance. Specifically, for a pre-trained weight matrix $W_0 \in \mathbb{R}^{d \times k}$, LoRA introduces two trainable matrices $A \in \mathbb{R}^{d \times r}$ and $B \in \mathbb{R}^{r \times k}$, where r is the adaptation rank and being much smaller than both d and k . Instead of directly fine-tuning W_0 ,

the model learns the update ΔW as follows:

$$W = W_0 + \Delta W = W_0 + A \times B \quad (1)$$

2.2. Multitask memory LoRA

In prior methods for downstream task fine-tuning, a task-specific LoRA matrix is appended to each layer of a large language model. Although this method is effective for individual downstream tasks, employing a shared LoRA matrix across all tasks can limit task interaction and exacerbate conflicts between tasks during multitask inference. This limitation reduces the ability of model to leverage cross-task information and address task-specific conflicts, thereby impeding overall multitask performance. To simultaneously handle multiple tasks and enhance the performance of subtasks, we propose a multitask memory parameter-efficient fine-tuning method for multimodal SER, which is designed to improve multitask fine-tuning capabilities. The proposed MMLoRA is illustrated in Figure 1.

The MMLoRA framework introduces several pivotal components: a universal LoRA expert, routing networks, a task-specific mixture of LoRA experts, and task memory units, with the trainable elements being the routing networks and the LoRA experts. It should be noted that different tasks have their own task-specific memories, and multiple task objectives achieve cross-layer information transmission through their respective memory units. Consider the main task as an example: in this case, the memory LoRA of the current layer receives the hidden state from the general output of the previous layer, which is then processed through the task-specific mixture of LoRA experts. Finally, the main task memory unit in the current layer is updated by integrating the main task memory from the previous layer, general task-related information, and the task-specific details of the current layer.

Specifically, the input token x_i is first processed by the routing network $R(\cdot)$, which is a linear layer that computes the probability distribution P_N of assigning the input token x_i to

each LoRA expert or task-specific memory:

$$P_N = \text{softmax}(R(x_i)) \quad (2)$$

where N represents the total number of LoRA experts and the task memory unit. The task-specific mixture of LoRA experts includes $N - 2$ experts, from which the top-K experts are selected based on the probabilities computed by the routing network:

$$P_{N-2}^{(k)} = \text{top-K}(P_N) \quad (3)$$

The task memory unit is a key component of MMLoRA. It is composed of three main elements: the universal LoRA expert $\mathcal{E}_u$, the top-K experts $\mathcal{E}_k$ from the task-specific mixture of experts, and the task memory M_{l-1} from the previous layer. The universal LoRA expert captures shared task information, while the task-specific experts handle task-specific information. The routing network assigns a probability P_u to the universal LoRA expert, $P_K^{(j)}$ to each of the top-K task-specific experts, and P_m to the task memory from the preceding layer. The task memory at the current layer l is then computed as follows:

$$M_l = P_u \mathcal{E}_u(x_i) + \sum_{j=1}^K P_K^{(j)} \mathcal{E}_k(x_i) + P_m M_{l-1} \quad (4)$$

This task memory mechanism allows the model to retain and propagate task-specific information across layers, thereby facilitating cross-task interaction during inference. As illustrated in Figure 1(b), the final layer differs from the previous layers as the task-specific memory for each task is directly added to the final output to generate task-specific head. Thus, the MMLoRA enhances simultaneous multitask inference by improving task interaction through parameter-efficient fine-tuning.

2.3. Fine-tuning for gender-assisted multimodal SER

The pipeline for gender-assisted multimodal SER based on VideoLLaMA is illustrated in Figure 2. During the fine-tuning phase, the LLM is carefully trained to simultaneously understand both emotion and gender, where gender recognition serves to enhance emotion recognition. The MMLoRA is added into the FFN layers of the LLM module within Video-LLaMA. Consequently, two task-specific outputs are generated in the last layer. The original LM head is replaced with two classification heads, and the training objective is to minimize the sum of the cross-entropy losses associated with both classification tasks. The loss function is defined as follows:

$$\mathcal{L} = - \sum_i (y_i^{(e)} \log \hat{y}_i^{(e)} + y_j^{(g)} \log \hat{y}_j^{(g)}) \quad (5)$$

where $y_i^{(e)}$ and $\hat{y}_i^{(e)}$ represent the true and predicted labels for emotion, respectively, while $y_j^{(g)}$ and $\hat{y}_j^{(g)}$ correspond to the true and predicted labels for gender.

To construct the instruction set for fine-tuning, assume a dataset containing N conversation instances. The i -th conversation C_i comprises K multimodal utterances, represented as $C_i = \{(h_k, M_k), Y_k\}$, where h_k denotes the historical dialogue text, M_k represents the k -th target utterance to be classified, and Y_k is the corresponding emotion label, with $i \in [1, 2, \dots, N]$ and $k \in [1, 2, \dots, K]$. Each target utterance M_k consists of both text (T) and speech (V) modalities, i.e., $M_k = (T_k, A_k)$, where the audio tokens A_k are projected into the embedding space of the LLM using

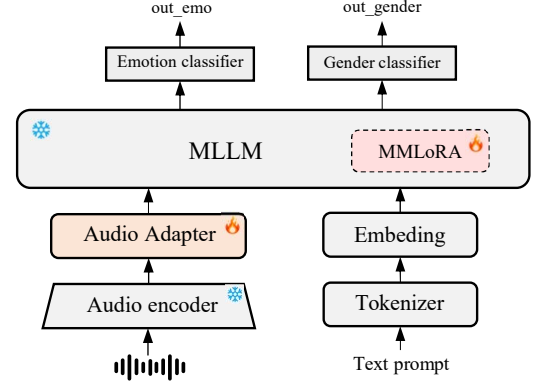


Figure 2: *The pipeline of gender-assisted multimodal SER based on VideoLLaMA. Only the MMLoRA module and the audio adapter, consisting of the Q-former and a linear layer, are trainable.*

the audio encoder and audio adapter from Video-LLaMA. Instructions are generated according to predefined task specifications. For instance, a prompt template may be as follows: "Context: [contextual_history]. Audio tokens: [audio_tokens]. Utterance: [current_utterance]. Given the context, audio tokens, and utterance, determine both the emotion and gender of the current speaker simultaneously.". Subsequently, the trainable parameters are updated by optimizing the multitask loss.

3. Experiments

3.1. Dataset

The dataset used for our experiments is the IEMOCAP [18], a widely recognized dataset for speech emotion recognition. We utilized a subset of IEMOCAP, comprising 5,531 utterances categorized into four emotion classes: angry, happy (with excited relabeled as happy), sad, and neutral. The experiments were conducted using a leave-one-session-out cross-validation approach.

3.2. Multimodal large language model

Video-LLaMA is an open-source multi-modal framework that enables LLM to simultaneously process both the visual and auditory content and engage in conversation with humans. In this paper, we use a modified version of Video-LLaMA, with the visual branch removed and utilize the audio-language branch of Video-LLaMA, which includes a pre-trained ImageBind [19] as the audio encoder. The process begins by converting the audio clips into spectrograms using Mel-spectrogram bins. The audio encoder then maps each spectrogram into a dense vector. This branch also features a positional embedding layer to inject temporal information, an audio Q-former that fuses features from different audio segments, and a linear layer that maps the audio representations into the embedding space of LLMs. The version of Video-LLaMA used in our experiments is Video-LLaMA-2-13B-Pretrained.

3.3. Experimental Setup

During the training phase, the LLaMA2 component of VideoLLaMA is kept frozen, and only the parameters of the MMLoRA-related modules and the audio adapter are optimized using

AdamW [20] with a learning rate of $3e-4$. The model is trained for 10 epochs. Regarding the hyperparameter settings of MMLoRA, the task-specific mixture of LoRA experts includes 8 experts, with the top-K routing value set to 2. Experiments were conducted on a system equipped with NVIDIA A800 80GB GPUs. For evaluation, we used the widely adopted weighted accuracy (WA) and unweighted accuracy (UA) metrics for the IEMOCAP dataset.

3.4. Experiment Results and Analysis

To verify the effects of the proposed MMLoRA method, we conduct ablation experiments. First, we evaluated the multimodal emotion understanding capabilities of Video-LLaMA, and conducted experiments using both zero-shot and few-shot inference methods. In these experiments, we adopted more stringent instruction settings to ensure that the model could produce the expected results. For instance, when evaluating on the IEMOCAP dataset, we specified instructions such as: *"Given the context the context, audio tokens, and utterance, detect the emotion of the current speaker. The emotion must be one of the following: ['neutral', 'happy', 'anger', 'sad']. The output must be formatted exactly as follows: [one emotion]."* For the few-shot experiment, we provided three examples to guide the model. These examples helped the model understand the output format and improve its emotional classification by showing how different emotions are expressed in various contexts. The results of the zero-shot and few-shot experiments are shown in Table 1. The few-shot performance on the IEMOCAP dataset improved by over 10% compared to the zero-shot setting, demonstrating the effectiveness of example guidance in improving emotion classification accuracy.

The proposed MMLoRA is a parameter-efficient method designed for multitask learning, specifically aimed at enhancing the performance of emotion recognition subtasks. To evaluate its effectiveness, we compared the downstream emotion recognition performance of vanilla LoRA under the same MLLM framework. To ensure a fair comparison, the rank parameters and training hyperparameters of LoRA were kept identical to those of MMLoRA, with both methods applied to the FFN layers. As shown in Table 1, the proposed MMLoRA achieves 81.11% WA and 82.13% UA on IEMOCAP, which indicate that MMLoRA produces a better results based on multitask memory design.

To validate the effectiveness of the MMLoRA architecture under the same multi-task conditions, we conducted an ablation study using Multi-task LoRA. Unlike the multi-output of MMLoRA, this variant employs a single output supervised by the losses of both tasks simultaneously. This experiment aims to evaluate the advantages of structures such as memory units over the vanilla LoRA architecture in leveraging multi-task learning

Table 1: The performances of different methods with VideoLLaMA on IEMOCAP.

Methods	IEMOCAP	
	WA(%)	UA(%)
Zeroshot	45.77	47.11
Fewshot	58.36	57.73
LoRA	79.72	80.75
Multi-task LoRA (w/o memory)	79.93	80.22
MOE-LoRA (w/o gender)	80.02	80.97
MMLoRA	81.11	82.13

to enhance downstream task performance. For a fair comparison, both methods were configured with the same rank parameters and training hyperparameters. As shown in Table 1, the results highlight the effectiveness of MMLoRA in improving performance through multi-task learning.

To further validate the architectural design choices, we developed MOE-LoRA by removing both the task memory unit and universal LoRA expert from MMLoRA, resulting in a pure mixture-of-experts variant. This ablation necessitates single-task training without gender information integration, leading to performance degradation. This experiment further underscores the benefits of incorporating the task memory and universal LoRA expert in MMLoRA, as their inclusion introduces gender information, thereby enhancing emotion recognition by better accommodating the influence of gender differences. The results of the ablation experiment highlight the importance of gender task assistance in MMLoRA.

3.5. Comparison to previous methods

Table 2 presents the performance comparison between the proposed MMLoRA method based on VideoLLaMA and several previously published emotion recognition systems on the IEMOCAP dataset. The selected methods include unimodal approaches (audio-only) and multimodal approaches (audio and text), as well as a method utilizing large language models (LLMs) based solely on the text modality [21]. As shown in the table, the MMLoRA method achieves competitive results, particularly outperforming previous multimodal approaches. These findings highlight the effectiveness of incorporating gender-assisted multitask learning into large-model fine-tuning strategies. The MMLoRA enhances the ability of MLLM to generalize across diverse emotional expressions, resulting in improved performance of multimodal SER.

Table 2: Results of the MMLoRA-VideoLLaMA and previously published models on the IEMOCAP dataset.

Methods	Modality		WA(%)	UA(%)
	Audio	Text		
[Fang et al.,2023] [22]	✓		74.03	74.95
[Shen et al.,2024] [23]	✓		72.43	73.88
[Feng et al.,2024] [21]		✓	-	60.19
[Sun et al.,2023] [24]	✓	✓	78.42	79.71
[Kyung et al.,2024] [25]	✓	✓	76.11	77.16
[Wang et al.,2024] [26]	✓	✓	77.88	77.44
MMLoRA (ours)	✓	✓	81.11	82.13

4. Conclusion

In this paper, we introduced a novel multitask memory parameter-efficient fine-tuning method designed to enhance multimodal speech emotion recognition by incorporating gender as an auxiliary task. MMLoRA leverages both shared and task-specific LoRA experts, as well as a memory mechanism, to enable efficient multitask learning and improve generalization across diverse emotional expressions. The experimental results on the IEMOCAP dataset demonstrated that MMLoRA outperforms existing methods, providing a robust and scalable solution for parameter-efficient fine-tuning in emotion recognition task. These findings underscore the value of the proposed framework and offer valuable insights that may extend to broader applications of multitask learning within large model fine-tuning contexts.

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6. References

- [1] R. W. Picard, *Affective computing*. MIT press, 2000.
- [2] B. Schuller, R. Müller, M. Lang, and G. Rigoll, “Speaker independent emotion recognition by early fusion of acoustic and linguistic features within ensembles,” in *INTERSPEECH*. ISCA, 2005, pp. 805–808.
- [3] Y. Fang, X. Xing, Z. Chu, Y. Du, and X. Xu, “Individual-aware attention modulation for unseen speaker emotion recognition,” *IEEE Transactions on Affective Computing*, 2024.
- [4] L. Zhang, L. Wang, J. Dang, L. Guo, and Q. Yu, “Gender-aware cnn-blstm for speech emotion recognition,” in *Artificial Neural Networks and Machine Learning–ICANN 2018: 27th International Conference on Artificial Neural Networks, Rhodes, Greece, October 4-7, 2018, Proceedings, Part I 27*. Springer, 2018, pp. 782–790.
- [5] Y. Li, T. Zhao, T. Kawahara *et al.*, “Improved end-to-end speech emotion recognition using self attention mechanism and multitask learning,” in *Proc. INTERSPEECH 2019*, 2019, pp. 2803–2807.
- [6] C. Le Moine, N. Obin, and A. Roebel, “Speaker attentive speech emotion recognition,” in *Interspeech 2021*. ISCA, 2021, pp. 2866–2870.
- [7] W. X. Zhao, K. Zhou, J. Li, T. Tang, X. Wang, Y. Hou, Y. Min, B. Zhang, J. Zhang, Z. Dong, Y. Du, C. Yang, Y. Chen, Z. Chen, J. Jiang, R. Ren, Y. Li, X. Tang, Z. Liu, P. Liu, J.-Y. Nie, and J.-R. Wen, “A survey of large language models,” *arXiv preprint arXiv:2303.18223*, 2023. [Online]. Available: <http://arxiv.org/abs/2303.18223>
- [8] H. Touvron, L. Martin, K. Stone, P. Albert, A. Almahairi, Y. Babaei, N. Bashlykov, S. Batra, P. Bhargava, S. Bhosale *et al.*, “Llama 2: Open foundation and fine-tuned chat models,” *arXiv preprint arXiv:2307.09288*, 2023.
- [9] J. Bai, S. Bai, Y. Chu, Z. Cui, K. Dang, X. Deng, Y. Fan, W. Ge, Y. Han, F. Huang *et al.*, “Qwen technical report,” *arXiv preprint arXiv:2309.16609*, 2023.
- [10] H. Zhang, X. Li, and L. Bing, “Video-llama: An instruction-tuned audio-visual language model for video understanding,” *arXiv preprint arXiv:2306.02858*, 2023.
- [11] H. Liu, C. Li, Q. Wu, and Y. J. Lee, “Visual instruction tuning,” *Advances in neural information processing systems*, vol. 36, 2024.
- [12] J. Achiam, S. Adler, S. Agarwal, L. Ahmad, I. Akkaya, F. L. Aleman, D. Almeida, J. Altschmidt, S. Altman, S. Anadkat *et al.*, “Gpt-4 technical report,” *arXiv preprint arXiv:2303.08774*, 2023.
- [13] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, “LoRA: Low-rank adaptation of large language models,” in *International Conference on Learning Representations*, 2022.
- [14] Q. Liu, X. Wu, X. Zhao, Y. Zhu, D. Xu, F. Tian, and Y. Zheng, “Moelora: An moe-based parameter efficient fine-tuning method for multi-task medical applications,” *arXiv preprint arXiv:2310.18339*, 2023.
- [15] D. Li, Y. Ma, N. Wang, Z. Cheng, L. Duan, J. Zuo, C. Yang, and M. Tang, “Mixlora: Enhancing large language models fine-tuning with lora based mixture of experts,” *arXiv preprint arXiv:2404.15159*, 2024.
- [16] S. Chen, Z. Jie, and L. Ma, “Llava-mole: Sparse mixture of lora experts for mitigating data conflicts in instruction finetuning mllms,” *arXiv preprint arXiv:2401.16160*, 2024.
- [17] Z. Song, J. Zhuo, Y. Yang, Z. Ma, S. Zhang, and X. Chen, “Lora-whisper: Parameter-efficient and extensible multilingual asr,” in *Interspeech 2024*, 2024, pp. 3934–3938.
- [18] C. Busso, M. Bulut, C.-C. Lee, A. Kazemzadeh, E. Mower, S. Kim, J. N. Chang, S. Lee, and S. S. Narayanan, “Iemocap: Interactive emotional dyadic motion capture database,” *Language resources and evaluation*, vol. 42, pp. 335–359, 2008.
- [19] R. Girdhar, A. El-Nouby, Z. Liu, M. Singh, K. V. Alwala, A. Joulin, and I. Misra, “Imagebind: One embedding space to bind them all,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 15 180–15 190.
- [20] I. Loshchilov, “Decoupled weight decay regularization,” *arXiv preprint arXiv:1711.05101*, 2017.
- [21] T. Feng and S. Narayanan, “Foundation model assisted automatic speech emotion recognition: Transcribing, annotating, and augmenting,” in *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2024, pp. 12 116–12 120.
- [22] Y. Fang, X. Xing, X. Xu, and W. Zhang, “Exploring downstream transfer of self-supervised features for speech emotion recognition,” in *Proc. INTERSPEECH*, vol. 2023, 2023, pp. 3627–3631.
- [23] S. Shen, Y. Gao, F. Liu, H. Wang, and A. Zhou, “Emotion neural transducer for fine-grained speech emotion recognition,” in *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2024, pp. 10 111–10 115.
- [24] D. Sun, Y. He, and J. Han, “Using auxiliary tasks in multimodal fusion of wav2vec 2.0 and bert for multimodal emotion recognition,” in *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2023, pp. 1–5.
- [25] J. Kyung, S. Heo, and J.-H. Chang, “Enhancing multimodal emotion recognition through asr error compensation and llm fine-tuning,” in *Interspeech 2024*, 2024, pp. 4683–4687.
- [26] Y. Wang, D. Li, and J. Shen, “Inter-modality and intra-sample alignment for multi-modal emotion recognition,” in *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2024, pp. 8301–8305.